RFQ-HWP

Dual half-wave plate for 405 and 810 nm

Technical quotation request | Revision 0.1 | 2026-11-05

Supplier	Candidate	Quantity
Newlight Photonics sales@newlightphotonics.com	WPD03-H810-H405	1 + optional spare

Please quote quantity 1 plus an optional separately priced spare. This is a technical and budgetary RFQ, not a purchase order. No manufacturing release is implied.

Requester: Thomas Gfeller, X6 photonics project, Switzerland. Please return the quotation to the sender. Delivery address and billing details will be supplied with any later order.

Requirements and evidence

RFQ-HWP-01 Required retardance

Half-wave at BOTH405 and 810 nm: $\delta(\lambda)=\pi$ modulo 2π. WPD03-H810-F405 is not an equivalent substitution.

Please provide: Confirm exact order code and retardance at both wavelengths.808/404 variants require a new fit to actual wavelengths.

RFQ-HWP-02 Accuracy and band

Project target nearest-half-wave phase error$\leq 2\pi/100$ radians ($\lambda/100$ retardance) at each operating band, not merely at one catalog center.

Please provide: Guaranteed error and measurement uncertainty at actual wavelengths; provide $\delta(\lambda)$ over 400-410 and 800-820 nm where available, with temperature/AOI stated. Spans are characterization requests, not frozen passbands.

RFQ-HWP-03 Orders and axes

Nominal fast-axis angle 45° relative to local H/V. The plate rotates the pump before generation in one direction and the pair after generation in the other.

Please provide: Unwrapped retardance/order integers, fast/slow-axis identification at both wavelengths, marking tolerance and a clearly defined viewing direction.

RFQ-HWP-04 Bidirectional transfer

Reverse traversal must be included in the source Jones model. Forward intensity swapping is not sufficient.

Please provide: Forward/reverse Jones description, axis convention and any asymmetric/coating phase; state reciprocity assumptions explicitly.

RFQ-HWP-05 Temporal distinguishability

The two photons encounter different polarization histories. A half-wave value modulo 2π does not specify differential delay.

Please provide: $d\delta/d\omega$ or group-delay difference over 810 band, temperature dependence and thickness/material information sufficient for an independent estimate.

RFQ-HWP-06 Geometry

Catalog candidate: optical diameter 15 mm in 25.4-mm holder. Current worksheet assumes 1 mm optical material, not verified.

Please provide: Actual number/material/thickness of plates and air gaps; thickness tolerances, wedge, clear aperture, holder thickness, optical-center datum and CAD.

RFQ-HWP-07 AR and transmission

AR at 405 and 810 for incidence nominal 0°; sufficient clear aperture for the complete beam and adjustment range.

Please provide: Per-face R and total T versus wavelength and polarization, scatter, wavefront and fluorescence data; confirm family specification applies to offered unit.

RFQ-HWP-08 Angular/thermal response

Laboratory use; adjacent crystal oven 20-60 °C does not define the plate temperature. Small alignment tilt must be budgeted.

Please provide: δ sensitivity to angle and temperature, recommended operating limits and axis stability; propose zero-order alternative if bandwidth demands it.

RFQ-HWP-09 Mount and calibration

One adjustable manual holder for this plate. Nominal 45° is a start setting only.

Please provide: Compatible holder part number and envelope; recommended polarization/retardance calibration and marking inspection.

RFQ-HWP-10 Documentation

Supplier data must separate guaranteed values from typical or calculated values.

Please provide: Serial/lot certificate, spectral data in CSV, drawings, part revision and acceptance/remedy terms. Price any selected-performance option separately.

Quotation and response format

State unit price, currency, NRE, minimum order, lead time from approved specifications, quotation validity, warranty and shipping/tax terms separately. Identify included mounts, controllers and certificates to avoid double counting.

For each requirement ID, return COMPLY / DEVIATION / NOT AVAILABLE, the guaranteed value and conditions, and a document/curve reference. Highlight every exception; silence is not acceptance.

Data preferred: PDF datasheet/drawing, STEP geometry and CSV wavelength/temperature curves. Provide units, wavelength convention, polarization/port labels, measurement uncertainty and instrument resolution. Do not send proprietary information that must not be included in a public engineering record without flagging it.

Before an order: confirm the measured pump profile, final spectral bands, optical/mechanical fit and all agreed deviations in writing. Nominal 405/810 values permit preliminary quotation only.

The editable reply table is included in the workbook (Lieferantworten). No enquiry has been sent by preparation of this document.